
cLSTM: a Module for LSTM Architectures Classifying Visual Stimuli from Neural Patterns

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Abstract

This project proposes a LSTM module for a neural decoding task: predicting which natural image a mouse is viewing from its brain activity. Using the Allen Brain Observatory Neuropixels Visual Coding dataset, which records large-scale neural responses from 32 mice exposed to 118 images [AIBS, 2019], the task is framed as a 118-class time-series classification problem. Building on prior work that identified a single-layer LSTM as a strong baseline on this dataset [Carpenter et al., 2024], this report introduces a “Conscious LSTM” (cLSTM) that injects structured stochasticity into the hidden state via a learned low-rank covariance module inspired by quantum theories of consciousness. Extensive experiments across 96 training runs per model show that, under well-tuned hyperparameters, cLSTM yields an average test accuracy improvement of roughly 0.3% over the baseline LSTM across mice. The accuracy improvements with the current design are very modest and come at a high computational cost. The validity of the modeling hypothesis is unclear and requires further exploration.

1 Introduction

The ability to decode external visual stimuli solely from internal neural activity represents a significant frontier in both neuroscience and artificial intelligence. As recording technologies like Neuropixels probes [AIBS, 2019] evolve to capture large-scale spiking activity with high spatial and temporal resolution, the challenge shifts to developing computational models capable of interpreting this rich data. The goal of this project is to classify which of 118 natural images a mouse is viewing based on its neuronal firing rates, building on previous work that identified the best performing architecture.

Neural data is inherently high-dimensional and sequential, making Recurrent Neural Networks (RNNs) a natural fit for modeling. From biological understanding of information processing in animal brains, we know that a lot of information processing/encoding has already taken place, represented by the firing patterns of the measured neurons. Given this pre-processed information rich data, the challenge is to model how this information interacts and evolves over time, and finally translate that into 118 classification scores. Framed in this way, it would make sense that basic, temporally-informed neural networks like LSTM’s would perform well on this data. However, Neural networks assume deterministic relationships in the data, and thus are prone to overfitting, and may not be ideally suited to modeling potentially stochastic internal processes like consciousness, which likely affect neural interaction. Dropout helps by increasing flexibility, so that the model isn’t as reliant on specific, exact relationships it memorized from the data, but it is a very simplistic/blunt tool. It would perhaps be better to learn a structured way of injecting stochasticity into the model architecture to force it to be more flexible. This paper investigates whether attempting to model the probability distribution of the mice’s consciousness can enhance decoding performance. We propose the Conscious LSTM (cLSTM) module, which modifies the standard LSTM update rule by adding a state-dependent noise component with a learned distribution. This design allows the model to explore a probabilistic latent space, potentially capturing the variability and noise inherent in biological neural networks better than deterministic baselines.

1.1 Theories on Non-Deterministic Consciousness

Our architectural modifications are inspired by quantum consciousness theories. While traditional computational theories view consciousness as an emergent property of information processing, quantum consciousness hypothesizes that consciousness arises from physical, non-deterministic quantum phenomena. This suggests that stochastic processes are deeply involved in brain processes, rather than being purely deterministic.

2 Related Works

The interpretation of visual information from extensive neural recordings has emerged as a significant challenge in systems neuroscience, with immediate consequences for comprehending brain representations and advancing brain-machine interfaces. The visual cortex of a mouse serves as an exceptionally valuable experimental platform due to the accessibility of standardized datasets and high-resolution imagery.

[Xie et al., 2023] conducted research on the systematic evaluation of visual coding across various mouse brain regions and behavioral stages, utilizing Neuropixels datasets from the Allen Brain Institute. Through the utilization of linear decoders and generalized linear models, they demonstrated that the visual cortical and thalamic areas harbor the most robust stimulus information, whereas the hippocampus regions approach chance levels. Moreover, the study demonstrated that behavioral variability, such as locomotion, can substantially influence decoding accuracy.

[Schneider et al., 2023] expanded upon linear methodologies and presented CEBRA, a self-supervised contrastive embedding framework that generates coherent latent spaces from brain recordings. It attained a decoding capability surpassing 95% frame-by-frame accuracy on natural films in the mouse visual cortex. It also facilitated cross-species generalization and resilience across recording modes.

[Joel et al., 2025] have shown that natural movies can be reconstructed from calcium imaging data in mouse V1 utilizing dynamic neural encoding models. The results highlighted the significant impact of behavioral parameters, such running speed and pupil dilation, on decoding quality, indicating that considering state-dependent variability is essential for creating effective decoders.

These studies illustrate the potential and difficulties of brain decoding. Linear models offer interpretability, although they possess constrained flexibility. Recurrent and attention-based designs encapsulate temporal and relational structures, whereas contrastive learning methodologies facilitate cross-species generalizations.

2.1 Neural Decoding Baseline

Prior exploratory work ([Carpenter et al., 2024]) evaluated ten different machine learning architectures on this specific decoding task: Long Term Short Memory, Principal Component to MVLR, Transformer, Single Layer NN, Spatial Temporal GAT, Support Vector Classifier, Graph Attention Network, Multivariate Logistic Regression, Multi-layer Perceptrons, and Graph Convolutional Network (ordered from best performance to worst). [Carpenter et al., 2024] concluded that a simple, one-layer LSTM consistently outperformed more complex architectures, achieving the highest classification accuracy score (among the best performing subject mice of each model). Consequently, we adopt this high-performing one-layer LSTM as our primary baseline for benchmarking improvements.

3 Methodology

3.1 Dataset

We utilize data from the Allen Brain Observatory – Neuropixels Visual Coding experiment. The dataset comprises recordings from 32 mice, each subjected to a session where 118 natural images were flashed for 250 milliseconds, repeated 50 times per image [AIBS, 2019]. The spike activity was processed into time-binned firing rate counts (5 bins per 250 millisecond training example), resulting in ordered a time-series where each row corresponds to one time bin, grouped in blocks of 5

corresponding to one training example. Each column except for the last represents a 'unit', and the last column corresponds to the associated frame (numerically encoded). The physical probes into the mice's brain are not precise enough to measure individual neurons, but rather they collect data on brain regions from which individual neuron spike activity is mathematically derived based on unique waveform signatures [AIBS, 2019]. This two dimensional dataset is split into predictor data and labels (X and y), and X is reshaped into a 3-dimensional tensor with dimension:

$$(\# \text{ image exposure (samples)} \times \# \text{ time bins (5)} \times \# \text{ neurons (varies per mouse)}) \quad (1)$$

While y remains a one-dimensional label vector. The reshaped dataset is then split into 72% training, 20% test, and 8% validation, using stratification. Crucially, the model has to be trained on each mouse separately to account for inter-subject variability in neural topology.

3.2 Baseline LSTM Architecture

The baseline model is a standard one-layer Long Short-Term Memory (LSTM) network, with hidden state dimension of 1000 [Carpenter et al., 2024]. For each time bin, It combines and processes that bin's firing rate counts for each neuron (features), along with the previous LSTM unit (from previous time step, same network)'s cell ($c_t - 1$ and hidden state ($h_t - 1$), through the Forget, Input, Candidate Memory, and Output gates. The new $c_t - 1$ and $h_t - 1$ are outputted. These state are then passed into the next LSTM Unit which combines it with the next time bin, and so on until the final hidden state, which is passed into a fully connected output classification layer with dimension equal to the number of classes (118). The objective function uses Cross Entropy Loss between the final layer and the label vector y , and the ADAM Optimizer with default parameters.

3.3 Conscious LSTM (cLSTM) Module Architecture

We attempt to model non-deterministic consciousness by modifying the hidden state h_t between LSTM unit time step transitions with a random vector sampled from a multivariate distribution with a learned covariance matrix which is conditioned on the network's current state (during training). During evaluation, this random vector is set to 0 for more consistent validation/test accuracy scores. The modified $\tilde{h}_t$ is passed on to the next LSTM unit instead of h_t . Formally:

$$\tilde{h}_t = h_t + c \cdot A(h_t)z \quad (2)$$

Where:

- $h_t \in \mathbb{R}^d$ is the deterministic LSTM hidden state.
- $z \sim \mathcal{N}(0, I_k)$ is a standard normal noise vector of dimension k .
- $A(h_t) \in \mathbb{R}^{d \times k}$ is a learned matrix defining the covariance structure $\Sigma(h_t) = A(h_t)A(h_t)^\top$.
- $c \in [0, 1]$ is a hyperparameter scaling coefficient controlling the magnitude of the stochastic injection.

Due to the fact that we cannot sample from $\mathcal{N}(0, \Sigma(h_t))$ directly, we use the following reparameterization trick:

want to sample: $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma)$

given: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$, $\Sigma = AA^\top$

then: $\mathbf{x} = A\mathbf{z}$

Which ensures the network remains differentiable throughout backpropagation. $A(h_t)$ is low rank, the size of which is controlled by hyperparameter k .

The matrix $A(h_t)$ is computed by a dedicated fully connected network (the "A-network") which takes h_t as input and compresses it through two hidden layers (reducing dimensions by half at each step) before outputting the $d \times k$ entries, the vectorized view of $A(h_t)$. This allows the model to learn a state-dependent noise covariance, effectively reshaping the "cloud" of probable hidden states based on the current context. Since $A(h_t)$ is designed to model the probability distribution of the

Mouse’s ‘free will’, we want it to be influenced by the mouse’s current internal state rather than being independent of it, which we feel is a more realistic representation.

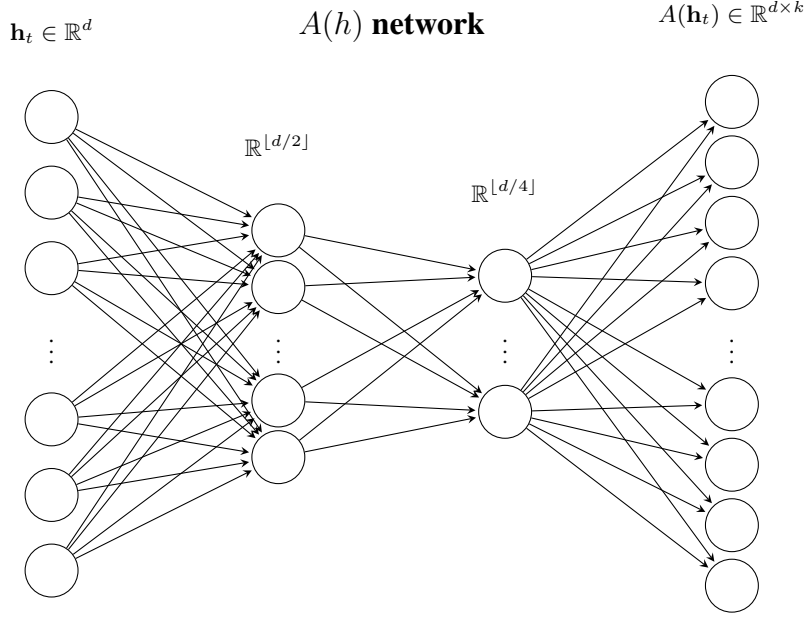


Figure 1: Neural network used to compute $A(\mathbf{h}_t)$.

4 Results

4.1 Hyperparameter Tuning Experimental Setup

The baseline LSTM architecture was tuned in the original paper. Our only modification was changing epochs from 15 to 9 because we noticed that performance plateaued and sometimes decreased after 9. For the cLSTM, we performed grid search starting with the same hyperparameters as the baseline, exploring many different combinations. We eventually settled on only modifying h_t dimension from 1000 to 2000, in addition to tuning k and c .

Table 1: Optimal hyperparameters for the baseline and Conscious LSTM models.

Hyperparameter	Baseline LSTM	Conscious LSTM
d (hidden state dimension)	1000	2000
Number of LSTM layers	1	1
Sequence length	5	5
Batch size	64	64
Learning rate	0.001	0.001
Training Epochs	9	9
Optimizer	ADAM(default parameters)	ADAM(default parameters)
k ($A(h_t)$ rank)	N/A	30
c (scaling coefficient)	N/A	0.1

We noticed that increasing the number of LSTM layers beyond 1 consistently and significantly decreased performance, and once c was increased to about 0.6, performance on some mice dropped off significantly while others still saw performance gains. For these mice where performance decreased significantly at $c = 0.6$, validation accuracy peaked early and then dropped as epochs increased. Bin count wasn’t tuned robustly, but from limited trials we concluded that increasing to 10 decreased performance, while decreasing to 3 saw similar performance, which coincides with the results of []. ADAM Optimizer default parameters were not tuned, except for learning rate.

Once optimal hyperparameter values were determined, we trained both models 3 times on each of the 32 mice for a total of 96 training runs per model in order to account for high variability in training results due to the nature of the cLSTM design.

4.2 Performance Analysis

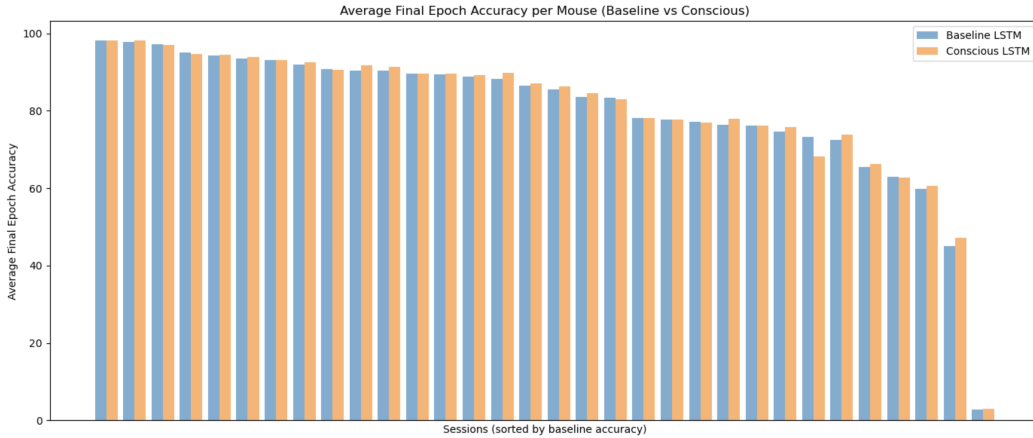


Figure 2: Test accuracy results.

Test accuracy gains were very modest, averaging about 0.3% across all 96 training runs. For most mice, differences between cLSTM and the baseline are practically undetectable, but for some mice improvements are noticeable, up to 2.13%. The average performance increase is brought down by the only mouse with a noticeable performance decrease of 5.01%.

Table 2: Computational Performance: baseline LSTM vs. cLSTM.

Model	Params	MACs (per fwd)	Avg runtime / session (s)	Avg peak GPU mem (MB)
Baseline LSTM	10,886,118	3,452,039,552	6.06	451.6
Conscious LSTM	62,337,618	19,905,351,552	60.31	1431.4
Conscious / Baseline	5.73×	5.77×	9.96×	3.17×

Parameter Count and MAC’s were calculated using the average neuron count across all mice. Runtime was also average across all mice, utilizing a NVIDIA A100 GPU. The increased computational costs of our cLSTM are significant.

5 Conclusion & Future Directions

While the performance gains are modest, the cLSTM framework opens several avenues for future investigation: (1) - Compare the complex learned covariance of the A-network against simpler regularization techniques like dropout to isolate the benefits of learned state-dependent noise. (2) - Quantify how heavily the downstream classifier depends on the stochastic perturbations versus the deterministic state. (3) - Redesigning the A-network for greater computational efficiency and performance. (4) - Implement the module in GRU rather than an LSTM, as over a short time frame with few bins, c_t seems unnecessary.

References

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